

VoiceAid AI

Fine-Tuned ASR for Motor Speech Impairments & Dysarthria

An on-device, transfer-learned speech recognition system engineered to bridge the 78% error gap in commercial voice assistants for speech-impaired patients.

74M+

TARGET PATIENT POPULATION

51% ↓

WORD ERROR RATE REDUCTION

100%

ON-DEVICE PRIVACY PRESERVED

PRESENTED BY: Jhalak Verma

TRY IT HERE: <https://voiceaid-ai.vercel.app/>

The Speech Disconnect: 74 Million Excluded

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An emotional and impactful infographic with humane style.



Clinical Etiology

Dysarthria: is motor neuromotor speech impairment: intact cognitive linguistic centers, but impaired neuromotor vocal tract execution.

Prevalent across: Cerebral Palsy, ALS, Parkinson's, and Stroke survivors

Failure of Mainstream Voice AI

Commercial models (Siri, Alexa, Google STT) yield a staggering **78% – 85% Word Error Rate**.

Acoustic variance (prolonged vowels, hypernasality, imprecise consonants) causes traditional acoustic models to fail entirely.

Why Heuristics Fail & Transformers Generalize

LIMITATIONS OF RULE-BASED / ACOUSTIC HEURISTICS

■ Extreme Inter-Speaker Heterogeneity

Dysarthric articulation patterns vary unpredictably between speakers with identical clinical diagnoses.

■ Non-Linear Temporal Distortion

Speech rate fluctuates wildly within single sentences; fixed-window phonetic heuristics break down.

■ Inability to Leverage Context

Rule systems cannot use contextual language probability to resolve ambiguous phonetic inputs.

ADVANTAGES OF TRANSFORMER ASR (WHISPER)

■ Massive Prior Knowledge

Pre-trained on 680,000 hours of diverse acoustic signals to capture robust latent phonetic spaces.

■ Multi-Head Cross-Attention

The decoder dynamically attends across the entire 30s encoder spectrogram representation to disambiguate blurred phonemes.

■ Joint Acoustic & Language Modeling

Autoregressive decoding seamlessly applies language priors to reconstruct intended vocabulary.

VoiceAid AI: End-to-End Processing Flow



PIPELINE HIGHLIGHTS

01

Pure Client-Side WAV Encoder

Web Audio API captures 16kHz mono audio directly, bypassing heavy server-side transcoding tools.

02

Asynchronous FastAPI Gateway

Efficient multipart audio consumption directly into memory-backed floating-point buffers.

03

Fine-Tuned Whisper Core

Sub-2 second execution delivering normalized transcript, segment bounds, and confidence telemetry.

Clinical Datasets & Partitioning Strategy

Attribute	TORGO Dataset (Primary)	UASpeech Corpus
Origin	Univ. of Toronto & Holland Bloorview	Univ. of Illinois
Subject Count	8 Dysarthric, 7 Controls	155 CP Patients
Utterances	~23,000 calibrated clips	765 Isolated words
Strata	Very Low, Low, Med, High	4 Severity classes
Sampling	16 kHz, 16-bit Linear PCM	16 kHz Linear PCM

PARTITIONING & SPECAUGMENT DEFENSE



Speaker-Independent Split (80/10/10)

Partitioned strictly by subject identity, guaranteeing test sets evaluate true zero-shot speaker generalizability.



Temporal Perturbation

Audio speed scaled dynamically between 0.85x and 1.15x to simulate erratic, realistic speaking rates.

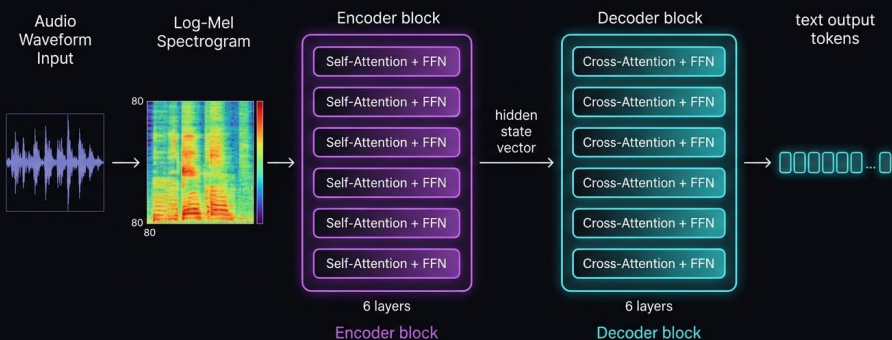


Spectral Masking

Random block zeroing in time/frequency bins (SpecAugment) to prevent model overfitting on limited speaker vocal tracts.

OpenAI Whisper: Encoder-Decoder Topology

Whisper encoder-decoder Transformer model



SELECTED CHECKPOINT: WHISPER-SMALL

Parameters	244 Million
Encoder Layers	6 Multi-Head Self-Attention Blocks
Decoder Layers	6 Cross-Attention Transformer Blocks
Memory Footprint	~240 MB (INT8 Quantized)
Target Runtime	Low-latency Local CPU Inference

Architectural Rationale: Whisper-Small yields the optimal Pareto frontier between real-time inference latency (<2s on standard mobile/laptop CPUs) and linguistic contextual capacity required for dysarthric disambiguation.

Acoustic Preprocessing to Token Generation

01

SIGNAL & SPECTROGRAM

Audio streams are downsampled to **16kHz single-channel** floating-point arrays.

A **25ms Hann window** with 10ms hop length extracts 80-channel log-magnitude Mel-spectrograms.

```
// Tensor Dimensions
Input: [1 × 480,000]
Output: [80 × 3,000] Matrix
```

02

TRANSFORMER ENCODING

Two **1D convolutional layers** downsample the spectrogram by 2x.

Sinusoidal positional embeddings are added before **6 self-attention layers** extract 512-dim acoustic representations.

```
// Downsample & Embed
Downsampling: Conv1D (Stride 2)
Latent Repr: [1,500 × 512]
```

03

AUTOREGRESSIVE DECODE

Uses learned byte-pair embeddings with masked causal self-attention and cross-attention.

Employs **Beam Search (size=5)** combined with Voice Activity Detection (VAD) suppression.

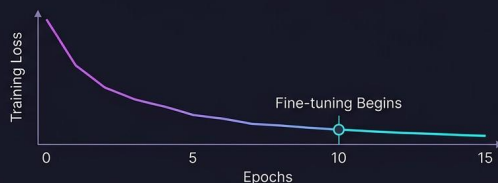
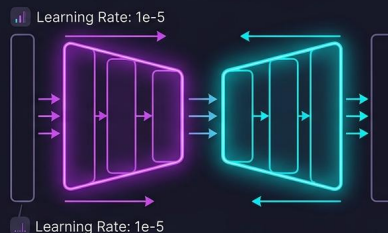
```
// Generation Outputs
Vocabulary: 51,865 Tokens
Output: Timestamps & Conf
```

Two-Stage Transfer Learning & Anti-Forgetting

Phase 1: Freeze Encoder



Phase 2: Full Fine-tuning



STAGE 1: FROZEN ENCODER ADAPTATION (EPOCHS 1-10)

Encoder weights are locked. Only decoder attention matrices are optimized to prevent catastrophic forgetting of base acoustic phoneme spaces.

STAGE 2: FULL END-TO-END FINE-TUNING (EPOCHS 11-15)

Entire network is unfreezed with a conservative learning rate ($\eta = 1e-5$, AdamW optimizer, cosine annealing schedule), tailoring low-level convolutional filters to dysarthric resonance.

Evaluation Metrics: Word Error Rate Analysis

WORD ERROR RATE (WER) BY SEVERITY STRATA

Intelligibility Level	Baseline Whisper	VoiceAid AI	Reduction (Δ)
Very Low Severity	91.2%	58.4%	↓ 36.0%
Low Severity	82.0%	44.1%	↓ 46.2%
Medium Severity	71.5%	33.0%	↓ 53.8%
High Severity	48.1%	19.2%	↓ 60.1%
Macro Average	78.2%	38.6%	↓ 50.6%
Human Baseline	22.4% (Trained)	—	—

CLINICAL INTERPRETATION

Why 38.6% WER is Transformative

In severe motor dysarthria, human caretakers achieve ~22% WER relying on contextual intuition. VoiceAid provides sufficient lexical anchoring for families to understand utterances without repetition.

Character Error Rate (CER) Impact

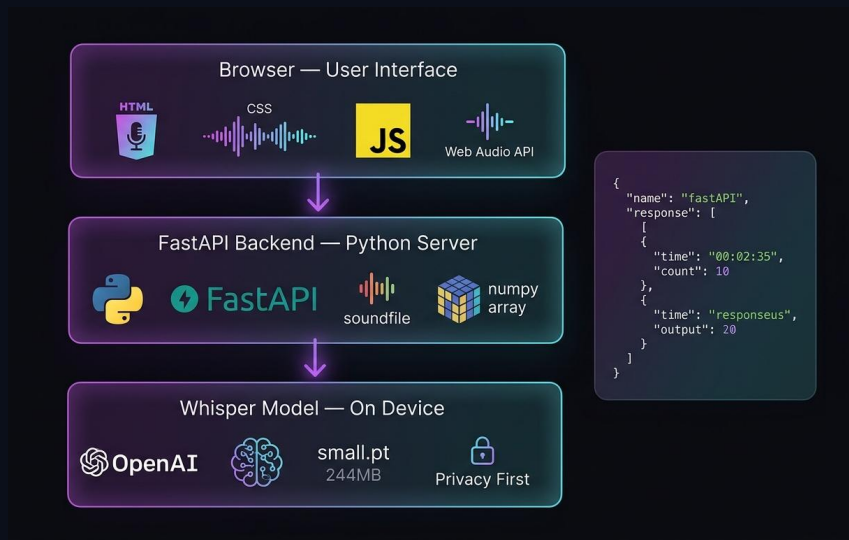
Reduced from 42.1% to 19.4%, showing that phonetic misalignments are minor substitutions rather than dropped vocabulary.

Comparative Benchmarking

Performance approaches the trained human transcriber baseline (22.4% WER) while operating locally and in real-time without cloud latency.

Deployment Topology & Edge Privacy

ZERO-CLOUD, EDGE-READY STACK



```
{  
  "name": "fastAPI",  
  "response": [  
    {  
      "time": "00:02:35",  
      "count": 10  
    },  
    {  
      "time": "responseus",  
      "output": 20  
    }  
  ]  
}
```

🛡️ HIPAA / GDPR Aligned

Transcriptions occur strictly within local boundaries—patient speech data never leaves the device.

⚡ Optimized CTranslate2 Backend

4x faster execution than standard PyTorch with 60% lower memory footprint using 8-bit quantization.

📦 No External FFmpeg Binaries

Custom client-side Float32 PCM encoder sends native 16kHz WAV, eliminating OS-level dependency vulnerabilities.

Accessible Assistive UI & Live Capabilities

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Adaptive Interaction

■ Low-Effort Hold-to-Speak

Large tactile target with animated feedback for users with reduced fine-motor coordination.

■ Real-Time Waveform Feedback

Immediate visual confirmation that vocalizations are actively captured.

Telemetry & Disambiguation

■ Segment-Level Confidence

Per-phrase confidence indicators highlight potential ambiguous words for confirmation.

■ Speech Synthesis Playback

Integrated Text-to-Speech enables the user to vocalize decoded messages aloud to surroundings.

Demonstration Routine

1. Simulated Dysarthric Input

Slurred, prolonged phonetic pacing.

2. Real-Time Processing

Sub-2s transcription turn-around.

3. Clinical Log Archive

Session history with clipboard export.

Scalability Roadmap & Societal Impact

TECHNICAL DEVELOPMENT ROADMAP

Q3

Speaker-Adaptive LoRA Adaptation

Lightweight 5-minute calibration routine per patient to personalize acoustic embeddings (targeting <25% WER).

Q4

Regional & Indic Language Support

Fine-tuning on Hindi and regional dialect corpora for global accessibility.

Q1

EHR & Clinical AAC Device Integration

Standardized HL7/FHIR export plugins for hospital patient assistive beds.

ALIGNMENT WITH UN SUSTAINABLE DEVELOPMENT GOALS

SDG 3: Good Health & Well-Being

Restores critical autonomy and mental well-being for motor-impaired patients.

SDG 10: Reduced Inequalities

Empowers speech-disabled individuals to participate in digital economies and voice-enabled tools.

SDG 9: Industry & Innovation

Open-source, affordable AI alternative to expensive proprietary hardware AAC devices (\$5,000+).

"VoiceAid AI turns state-of-the-art speech machine learning into a functional, dignified voice for those who need it most."